

Pulkit Juneja

DOCTORAL STUDENT

Texas A&M University, College Station

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[Personal Portfolio](#)

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[Google Scholar](#)

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Research Focus

My doctoral research revolves around remote sensing and process-based crop modeling in southern U.S. row crop systems. Current research pairs UAV LiDAR and hyperspectral imagery with machine learning to estimate canopy traits, and calibrate DSSAT and DayCent against field and flux-tower measurements to test management scenarios over long term decision making. Earlier work quantified cover crop effects on corn yield, nitrogen and soil greenhouse gas emissions.

Education

PhD in Agronomy

Texas A&M University

College Station, TX

Jan 2025 - Present

MS in Agronomy

Texas A&M University

College Station, TX

May 2022 - Dec 2024

BS (Hons) in Agriculture

Punjab Agricultural University

Ludhiana, India

Sept 2017 - July 2021

Experience

Graduate Research Assistant

Blackland Research & Extension Center, Texas A&M AgriLife Research

Temple, TX

May 2022 - Present

PhD Projects

- Managing three multi-year replicated field studies of 24–80 plots across four seasons, leading a crew of five through treatment application, in-season sampling, and data collection.
- Integrating UAV-based LiDAR and hyperspectral data to estimate corn leaf area index using machine learning; 12–15 flights across two seasons over 2+ acre research fields, with sensor integration reaching $R^2 = 0.75$ (RMSE = $0.41 \text{ m}^2 \text{ m}^{-2}$) within fields and LiDAR alone transferring better across fields.
- Evaluating a perennial ground cover (living mulch) with variable nitrogen rates to improve nitrogen retention and reduce soil erosion in corn production systems of the southern U.S.; crimson clover and fallow plots out-yielded white clover across nitrogen rates from 0 to 240 kg N ha^{-1} .
- Modeling net ecosystem exchange in a no-till sorghum-cotton rotation with the DayCent biogeochemical model, calibration and validation against eddy covariance flux tower measurements.
- Developed farmwth.com, a weather tool for Texas farmers and extension agents with 30 years of historical weather along with customizable analytics tailored to specific cropping practices; used by researchers and extension agents across about 20 counties.
- Simulating cover crop termination and cotton planting scenarios, to identify the window that protects cotton yield while securing the benefits of the cover crop.
- Assessing long-term impacts of conservation tillage on corn performance and soil health, with DSSAT modeling.

Master's Project

- Developed climate-smart agricultural recommendations using cover crops and crop residue management techniques; a U.S. meta-analysis showed cover crops increased corn yield by 11% and nitrogen content by 22%.
- Estimated greenhouse gas fluxes from corn (*Zea mays*) field using static chambers.
- Evaluated corn yield and water use efficiency following cover crop and variable nitrogen rate; sunn hemp produced higher corn grain and biomass than sorghum–sudangrass, with no difference in evapotranspiration.

Graduate Teaching Assistant

Department of Soil and Crop Sciences, Texas A&M University

College Station, TX

Fall 2024, Fall 2025

- Taught soil physical, chemical and biological properties to more than 100 undergraduate students in **Introduction to Soil Science (SCSC 301)** course about soil properties and their analysis through laboratory exercises and field visits.
- Assisted in grading, preparing instructional materials to support experiential learning.

Undergraduate Student Worker

Department of Soil Science, Punjab Agricultural University

Ludhiana, India

Sept 2021 - Dec 2021

- Assessment of nano nitrogen, zinc fertilizers applied to rice-wheat cropping system
- Analysed the accumulation of lead in rice and its effect on crop growth and vigour

Agronomy Intern (Practical Crop Production)

Department of Agronomy, Punjab Agricultural University

Ludhiana, India

June 2018 - Mar 2019

- Carried out cultivation of corn and mustard (*Brassica juncea*)
- Field scouting for crop establishment, disease, and pest infestation

Publications

- **Juneja, P.**, Baath, G.S., Brar, J.S., Flynn, K.C., Yost, J.L., & Rajan, N. (2026). Corn Responses to Nitrogen Fertilization as Influenced by Cover Cropping. A Meta-Analysis. *Agronomy for Sustainable Development*, 46(1), 3. (IF: 8.2) <https://doi.org/10.1007/s13593-025-01067-6>
- **Juneja, P.**, Baath, G.S., Brar, J.S., Yost, J.L., Flynn, K.C., & Rajan, N. (2026). Exploring the Potential of Warm-Season Cover Crops and Nitrogen Management for Improving Corn Yield and Water Use Efficiency. *Agrosystems, Geosciences & Environment*. (Under Review)
- Brar, J.S., Baath, G.S., **Juneja, P.**, Jeong, J., Yost, J.L., Flynn, K.C., & Wyatt, B.M. (2025). Impacts of Dairy Manure and Synthetic Fertilizers on Greenhouse Gas Emissions and Crop Yields: A Global Meta-Analytical Comparison. *Science of the Total Environment*, 1005, 180836. (IF: 8.0) <https://doi.org/10.1016/j.scitotenv.2025.180836>
- Flynn, K.C., Baath, G.S., Kolodziejczyk, C., **Juneja, P.**, Sapkota, B.R., Hajda, C., & Smith, D. (2026). UAV LiDAR Voxelization Enhances Estimation of Crop LAI Across Flight Altitudes. *Computers and Electronics in Agriculture*. (Under Review)
- **Juneja, P.**, Baath, G.S., Flynn, K.C., Kolodziejczyk, C., Sapkota, B.R., & Brar, J.S. (2026). Integrating Hyperspectral and LiDAR for Corn Leaf Area Index Estimation and Cross Field Generalization. (In Preparation)

Book Chapters

- Bawa, A., Baath, G.S., **Juneja, P.**, & Brar, J.S. (2024). Drone mapping for agricultural sustainability: applications and benefits. In D. Thakur, S. Kumar, H.A.S. Sandhu, & C. Prakash (Eds.), *Sustainable Development Using Geospatial Techniques* (pp. 411–436). Wiley. <https://doi.org/10.1002/9781394214426.ch16>

Technical Skills

Software/Programming Languages	Field Equipment	Lab Equipment
R, Python, SAS, JMP, JASP, Pix4D, ArcGIS Pro, ENVI, CloudCompare, Git, SQL, TypeScript (Next.js), DSSAT, DayCent	UAV (FreeFly Astro, DJI Phantom, Matrice 300 RTK), Eddy Covariance Flux Tower, LAI-2200C, Stevens soil moisture sensor, MC5 Geoprobe	Gas chromatograph-Trace 1300, NC Soil analyser, Flow Injection analyser

Certification: FAA Part 107 Remote Pilot Certificate (small unmanned aircraft systems)

Honors & Awards

Mentee- Bayer University Mentor Program (B4U)

Bayer Crop Science

Virtual

Feb 2025-Feb 2026

- Selected for the **B4U** (Bayer for University) professional mentorship program, paired with an industry scientist mentor to develop career goals and professional growth

Graduate Student Leadership Conference

- Selected to attend the leadership conference at the CANVAS (ASA-CSSA-SSSA) Annual Meeting

*Salt Lake City, UT
Nov-09-2025*

Corteva DELTA Scholar

Corteva Agriscience

*Indianapolis, IN
Aug-07-2023*

- Travel scholarship to present at Corteva DELTA symposium at Corteva headquarters.

2nd Place in Student Oral Presentation

Airborne and Satellite Remote Sensing: CANVAS (ASA-CSSA-SSSA) Annual Meeting

*Salt Lake City, UT
Nov-12-2025*

1st Place in Student Poster Presentation

Semi-Arid Dryland Cropping Systems: ASA, CSSA, SSSA International Annual Meeting

*San Antonio, TX
Nov-11-2024*

2nd Place in Student Poster Presentation

Global Climate Change Community: ASA, CSSA, SSSA International Annual Meeting

*St. Louis, MO
Nov-01-2023*

Leadership

Graduate Senator – Department of Soil and Crop Sciences

College Station, TX

Graduate and Professional Student Government, Texas A&M University

2025 - 2026

- Representative for the Department of Soil and Crop Sciences in the university's student government.
- Contribute to policy discussions and decision-making processes impacting graduate and professional students.
- Serve on the **Research Promotion Committee**, organizing invited talks, student competitions, and events that highlight and advance research across the university.

Graduate Student Ambassador

College Station, TX

Department of Soil and Crop Sciences, Texas A&M University

2025 - 2026

- Hosting prospective students, mentoring incoming students, representing the department at events.
- Raise student concerns at monthly departmental meetings to help improve the graduate experience.

Internal Programs Coordinator

College Station, TX

International Student Association, Texas A&M University

2025 - 2026

- Coordinate and facilitate officer meetings, overseeing internal organization and communication.
- Plan and execute one major event each semester to promote cultural engagement and student involvement.

State Secretary

Ludhiana, India

All India Agricultural Student Association

Jan 2022 - Dec 2022

- Established strategic partnerships with multiple agricultural companies, fostering invaluable research opportunities

Class Representative

Ludhiana, India

Class of 2021 (B.S. Agriculture)

2020 - 2021

- Liaison between student and faculty, raising student related issues at department level

Services

Reviewer

Agronomy Journal

3 manuscripts reviewed

(2025–Present)

Urban Agriculture and Regional Food Systems

1 manuscript reviewed

(2026–Present)

Session Moderator

Session Title: Airborne and Satellite Remote Sensing Oral II

Salt Lake City, UT

CANVAS (ASA-CSSA-SSSA) Annual Meeting

Nov-12-2025

Session Judge

Session Title: Agroclimatology and Agronomic Modeling Poster I

Salt Lake City, UT

Session Moderator

Session Title: Climate Change and Agriculture Oral II

San Antonio, TX

ASA, CSSA, SSSA International Annual Meeting

Nov-13-2024

Session Judge

Session Title: Airborne and Satellite Remote Sensing Oral I

San Antonio, TX

ASA, CSSA, SSSA International Annual Meeting

Nov-11-2024

Grant Proposals

Southern SARE (Sustainable Agriculture Research and Education) Graduate Student Grant - 2025

Understanding Yield and Water Use Dynamics in Corn Production Under Integrated Living Mulch Systems (\$22,000, not funded)

Southern SARE (Sustainable Agriculture Research and Education) Graduate Student Grant - 2024

Integrating Warm-Season Cover Crops in the Corn-based Cropping System of Central Texas: Nutrient Management, Water Use Efficiency, and Corn Yield (\$21,260, not funded)

Presentations

Oral Presentations

- **Juneja, P.**, Baath, G. S., Flynn, K. C., Brar, J. S., & Sapkota, B. R. (2025) Integrating Lidar and Hyperspectral Data with Machine Learning for Corn Leaf Area Index Estimation [Abstract]. CANVAS (ASA-CSSA-SSSA) 2025, Salt Lake City, UT. [Web link](#)
- **Juneja, P.**, Baath, G. S., Singh, K., Singh, H., Williams, D., & Dhillon, J. S. (2025) Evaluating the Influence of Living Mulch and Nitrogen Rate on Corn Performance in Southern United States. [Abstract]. CANVAS (ASA-CSSA-SSSA) 2025, Salt Lake City, UT. [Web link](#)
- **Juneja, P.**, Baath, G. S., Brar, J. S., Flynn, K. C., Yost, J. L., & Rajan, N. (2024) A Meta-Analysis to Understand the Combined Effect of Cover Cropping and Nitrogen Fertilization on Corn Yield in the U.S. [Abstract]. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX. [Web link](#)
- **Juneja, P.**, Baath, G. S., Brar, J. S., Yost, J. L., Flynn, K. C., Rajan, N., & Casey, K. D. (2023) Impact of cover crop management on greenhouse gas fluxes in corn production systems [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO. [Web link](#)

Poster Presentations

- **Juneja, P.**, Baath, G. S., Brar, J. S., Yost, J. L., Flynn, K. C., & Rajan, N. (2024) Corn Yield Response and Water Use Efficiency Following Warm-Season Cover Crops in Central Texas [Abstract]. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX. [Web link](#)
- **Juneja, P.**, Baath, G. S., Brar, J. S., Flynn, K. C., Yost, J. L., Rajan, N., & Casey, K. D. (2024) Greenhouse Gas Emissions from Corn Field as Influenced By Cover Crop Management [Abstract]. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX. [Web link](#)
- **Juneja, P.**, Baath, G. S., Brar, J. S., Yost, J. L., Flynn, K. C., & Rajan, N. (2023) Performance and water use efficiency of rainfed cover crop-corn production systems in central Texas [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO. [Web link](#)
- **Juneja, P.**, Baath, G. S., & Brar, J. S. (2023) Estimation of cover crop biomass using crop biophysical parameters. Corteva DELTA Symposium, Indianapolis, IN.
- **Juneja, P.**, & Baath, G.S. (2023) Influence of water stress on performance of three warm season legume species. Texas A&M Plant Breeding Symposium, College Station, TX.